CO2 - AT1

Analytical problem solving

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Course Name:- Computer Vision

1 Thresholding and intensity level slicing PCB inspection.

Given that:-

$$r = [25, 60, 95, 130, 165, 200, 237]$$

solder joint intensity band = 90-170.

a, Binary threshold transformation

Given that:-

$$s = \begin{cases} 255, & r \geq 128 \\ 0, & r < 128 \end{cases}$$

r	25	60	95	130	165	200	235
s	0	0	0	255	255	255	255

output:-

[0,0,0,255,255,255,255].

b intensity level slicing

Pixels blw 90 and 170 are changed to 255.

others remain unchanged.

* 20-25
* 60→60
* 95→255
* 130→255
* 165→255
* 200→200
* 235→235

Therefore [25,60,255,255,255,200,235]

c. Number and percentage of highlighted pixels.

Highlighted pixels are:-

95, 130, 165.

Number:- 3 pixels.

Total pixels = 7

$$\text{percentage} = \frac{3}{7} \times 100 = 42.86.$$

d. Mean Intensity:-

original Mean:-

$$\bar{r} = \frac{25+60+95+130+165+200+235}{7}$$

$$\bar{r} = 130.$$

Slice o/p mean:-

$$\bar{s} = \frac{25+60+255+255+255+200+235}{7}$$

$$\bar{s} = 183.57.$$

e. Appropriate transformation:-

"Intensity level slicing is more appropriate for isoating the solder joints because the required solder-joint range is specifically 90-170.

2 Piecewise linear Contrast stretching - X Ray image enhancement.

* Control points : $(r_1, s_1) = (70, 20)$

$(r_2, s_2) = (140, 230)$, $L = 256$

$r = [30, 70, 90, 110, 140, 200]$.

2 equations and slopes

segment 1: 0-70

points : $(0,0)$, $(70,20)$.

slope :- $M_1 = \frac{20-0}{70-0} = \frac{20}{70} = 0.2857$.

equation $\Rightarrow s = 857r$.

segment 2: 70-140

$M_2 = \frac{230-20}{140-70} = 3$.

$s - 20 = 3(r-70)$

$s = 3r - 190$.

segment 3: 140-255

points : $(40, 230)$, $(255, 255)$

$M_3 = \frac{255-230}{255-140} = 0.2174$

b. Transformed values

input	segment	outputs
30	0-70	8.57
70	70-140	20
90	70-140	80
110	70-140	140
140	70-140	230
200	140-255	243.04.

S = [8.57, 20, 80, 140, 230, 243.04].

c. width and Contrast gain

140 - 70 = 70.

So, original width = 70.

transformed width = 210.

Contrast gain: $G = \frac{210}{70}$

$\boxed{G = 3}$

d. Compression and expansion.

segment -1 slope = 0.2857 < 1 → intensity range

segment -2 slope = 3 > 1 → intensity range

is $\boxed{\text{expanded}}$.

2. Interpretation

The transformation improves visibility of the diagnostic region 70-140. because its intensity range.

3. spatial filtering - Impulse noise Removal

The 3x3 neighbourhood is

$$\begin{bmatrix} 52 & 55 & 58 \\ 54 & \boxed{255} & 57 \\ 53 & 56 & 59 \end{bmatrix}$$

255 = salt noise

a. 3x3 mean filter

$$\text{Mean} = \frac{52+55+58+54+255+57+53+56+59}{9}$$

$$\boxed{\text{Mean} = 77.67.}$$

b. weighted average filter.

$$\text{Mask} = \frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

weighted sum:-

$$\frac{1}{16}\left[52+2(55)+58+2(54)+4(255)+\right.$$

$$+2(57) + 53 + 2(56) + 59]$$

$$= \frac{1686}{16}$$

$$= 105.375 //$$

c Median filter:-

Average values in ascending order

52, 53, 54, 55, 56, 57, 58, 59, 255

The middle value is

Median filter = 56.

d Absolute error:-

$$R = \frac{52+55+58+54+57+53+56+59}{8}$$

$$\boxed{R = 55.5}$$

Mean filter error:-

$$[77.67 - 55.5]$$

$$22.17.$$

Weighted average filter Error:-

105.375 − 55.5

49.875

Median filter error

56 − 55.5

0.5

filter	output	Absolute error
Mean	77.67	22.17
weighted Avg	105.375	49.875
Median	56	0.5

2 Best filter:-

The median filter is the most appropriate

4 edge detection − sobel and prewitt Gradient Analysis.

Given region

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 90 \\ 10 & 90 & 90 \end{bmatrix}$$

a) sobel G_x and G_y

sobel G_x :- $G_x = (-1)(10) + 0(10) + 1(10) + (-2)(10) + 0(10) + 2(90) + (-1)(10) + 0(90) + 1(90)$

$$\boxed{G_x = 240}$$

sobel G_y :- $= (-1)(10) + (-2)(10) + (-1)(10) + 0 + 0 + 0 + (1)(10) + (2)(90) + 1(90).$

$$\boxed{G_y = 240}$$

b) Gradient Magnitude

Exact Magnitude:-

$$M = \sqrt{G_x^2 + G_y^2}$$

$$= \sqrt{(240)^2 + (240)^2}$$

$$\boxed{M = 339.41}$$

Approximation:- $M = |G_x| + |G_y|$

$$= 480.$$

% Error:-

$$\frac{480 - 339.41}{339.41} \times 100$$

$$= 41.42\ \%$$

c, Gradient Direction:-

$$\alpha = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

$$\boxed{\alpha = 45^\circ}$$

d, Prewitt operatoai-

Prewitt G_x:- $(-10+10)+(-10+90)+(-10+90)$

$$\boxed{G_x = 160}$$

Prewitt G_y:- $(-10-10-10)+(10+90+90)$

$$\boxed{G_y = 160}$$

Magnitude:-

$$M = \sqrt{(160)^2 + (160)^2}$$

$$M = 226.27$$

Approximation:-

$$M = 160+160$$

$$\boxed{M = 320}$$

% Error:-

$$\frac{320 - 226.27}{226.27} \times 100$$

41.42%

e Edge detection for $M \geq 250$

Sobel

$M_{sobel} = 339.41$

Since

$339.41 \geq 250$

Sobel detects an edge.

Prewitt:

$M_{prewitt} = 226.27$

$226.7 < 250$

Prewitt doesn't not detect edge.

5 Global thresholding - iterative and otsu segmentation:-

Intensity k	0	1	2	3	4	5	6	7
frequency k	3	6	4	1	1	5	8	4

Total pixels

$N = 32.$

a) Global Mean Intensity

$M_G = \frac{\Sigma r_k n_k}{N.}$

$$= \frac{0(3)+1(6)+8(4)+3(1)+4(1)+5(5)+6(8)+7(4)}{32}$$

$$\boxed{MG = T_0 = 3.8125}$$

b) Interactive global thresholding

Starting threshold: $T_0 = 3.8125$.

Mean of c1

$$M_1 = \frac{17}{14} = 1.2143.$$

Mean of c2

$$M_2 = \frac{105}{18} = 5.8333.$$

$$T = 3.5$$

Otsu's Method:-

$$\sigma_B^2(K) = P_1(K)\,P_2(K)\left[M_1(K) - M_2(K)\right]^2$$

Class $a = 0, 1, 2$

$$P_1 = \frac{13}{32} = 0.40625$$

$$P_2 = \frac{19}{32} = 0.5937$$

$$P_1 = \frac{14}{13} = 1.0769$$

$$m_2 = \frac{108}{19} = 5.6842.$$

Therefore:- $\sigma_B^2(2) = 5.1202.$

Maximum variance occurs at

$k=3$

d Binary mapping and object percentage

Intensity	Binary output
0	1 – Object
1	1 – Object
2	1 – Object
3	1 – Object
4	0 – Background
5	0 – Background
6	0 – Background
7	0 – Background.

Object pixels $= 3+6+4+1 = 14$.

e Comparison of two methods

Iterative threshold $T = 3.524$

Otsu threshold $k^* = 3$.

The Histogram is suitable for global thresholding.